

Introduction to Contract Theory and Auctions

More about revenues & Revenue Equivalence Theorem

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Expected Payment once again...

- Recall that in both auction formats, FPA and SPA, or for short I and II respectively, we have that the **interim expected payment** for each individual buyer is given by

$$m^A(v) = G(v) \times E[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v] \quad G \sim Y_1^{(n-1)}$$

$\frac{1}{G(v)} \int_0^v y g(y) dy = \int_0^v y g(y) dy$

where $A = I, II$.

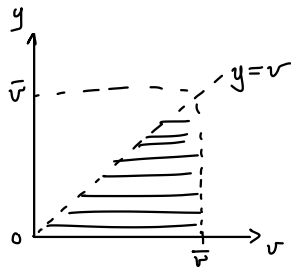
- Thus, the ex-ante payment for each individual buyer is given by

$$E[m^A(V)] = \int_0^{\bar{v}} m^A(v) f(v) dv$$

$$= \int_0^{\bar{v}} \left(\int_0^v y g(y) dy \right) f(v) dv$$

$$= \int_0^{\bar{v}} \underbrace{\left(\int_y^{\bar{v}} f(v) dv \right)}_{F(\bar{v}) - F(y)} y g(y) dy$$

$$= \int_0^{\bar{v}} (1 - F(y)) y g(y) dy$$



$$E[m^*(v)] = \int_0^v (1-F(y)) y g(y) dy$$

$$\begin{aligned} E[\text{Revenue}] &= n E[m^*(v)] = n \int_0^v (1-F(y)) y g(y) dy \\ &= \int_0^v \underbrace{n (1-F(y)) y g_1^{(n)}(y)}_{y g_2^{(n)}(y)} dy \\ &= \int_0^v y g_2^{(n)}(y) dy \\ &= E[\gamma_2^{(n)}] \end{aligned}$$

Auctioneer's Revenue once again. . .

- Note that the auctioneer's expected revenue in each auction format, R^A , is given by

$$\mathbb{E}[R^A] = n \mathbb{E}[m^A(V)] = n \int_0^{\bar{v}} y(1 - F(y))g_1^{(n-1)}(y)dy$$

Revenues from FPA vs. Revenues of SPA

- Although we know that $E[R^I] = E[R^{II}]$, it is not the case that R^I and R^{II} are the same random variable.

Example 6.1

Recall example 6.2, two bidders with $V_1, V_2 \sim U[0, 1]$ i.i.d. and $\beta^I(v) = \frac{v}{2}$ and $\beta^{II}(v) = v$. Then, by direct calculation:

$$\begin{aligned} G_2(y) &= F(y)^n + nF(y)^{n-1}(1 - F(y)) \\ &= nF(y)^{n-1} - (n-1)F(y)^n \end{aligned}$$

$$R^I = \max\{\beta^I(V_1), \beta^I(V_2)\} = \beta^I(\max\{V_1, V_2\}) = \beta^I(Y_1^{(2)}) = \frac{Y_1^{(2)}}{2}$$

$$Y_1^{(2)} \stackrel{\text{CDF}}{=} Y^2$$

$$Y_2^{(2)} \stackrel{\text{CDF}}{=} 2Y - Y^2 \sim R^{II} \quad R^{II} = \min\{V_1, V_2\} = Y_2^{(2)}$$

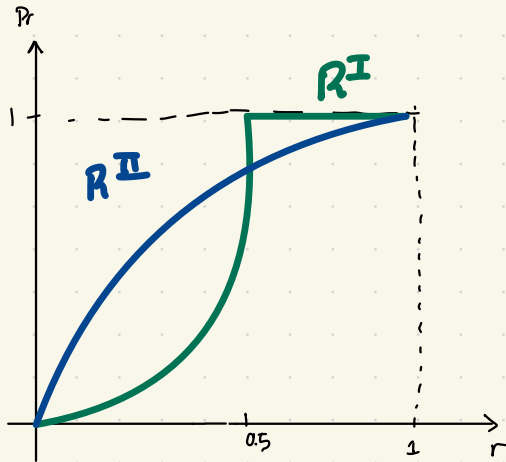
$$\begin{aligned} \text{CDF of } R^I : \Pr(R^I \leq r) &= \Pr\left(\frac{Y_1^{(2)}}{2} \leq r\right) = \underbrace{\Pr(Y_1^{(2)} \leq 2r)}_{\text{CDF of } Y_1^{(2)}} \\ &= (2r)^2 = 4r^2 \end{aligned}$$

$$R^I \stackrel{\text{CDF}}{\sim} 4r^2$$

$$R^{II} \stackrel{\text{CDF}}{\sim} 2r - r^2 = r(2-r)$$

$$\int_0^t 4r^2 dr$$

$$\int_0^t r(2-r) dr$$



- Let F_I be the distribution of R^I and F_{II} be the distribution of R^{II} .
- Example 7.1 suggests that F_{II} is a **mean-preserving spread** of F_I .
- This means that from the perspective of the auctioneer, the SPA is **riskier** than the FPA.
- Indeed, any **risk-averse auctioneer** would strictly prefer a first-price auction versus a second-price auction (assuming the bidders are risk-neutral).

Proposition 6.2

In the SIPV auction model, the distribution of equilibrium payments in a second-price auction is a mean-preserving spread of the distribution of equilibrium payments in a first-price auction. That is, F_I **second-order stochastically dominates** F_{II} .

Proof.

$$R^{\text{II}} = Y_2^{(n)}, \quad R^{\text{I}} = \beta(Y_1^{(n)}) \quad \beta(x) = E[Y_1^{(n-1)} | Y_1^{(n-1)} \leq x]$$

Goal is to show that R^{II} is a mean-preserving spread of R^{I}

$$\begin{aligned} E[R^{\text{II}} | R^{\text{I}} = r] &= E[Y_2^{(n)} | \beta(Y_1^{(n)}) = r] \\ &= E[Y_2^{(n)} | Y_1^{(n)} = \beta^{-1}(r)] \\ &= E[Y_1^{(n-1)} | Y_1^{(n-1)} < \beta^{-1}(r)] = \beta(\beta^{-1}(r)) = r \end{aligned}$$

$$\begin{aligned} \therefore E[R^{\text{II}} | R^{\text{I}} = r] = r &\Rightarrow \text{Construct } \varepsilon = R^{\text{II}} - E[R^{\text{II}} | R^{\text{I}}] \\ E[\varepsilon | R^{\text{I}}] = 0 &\Rightarrow R^{\text{II}} = R^{\text{I}} + \varepsilon \end{aligned}$$

The Revenue Equivalence Theorem

- Consider a SIPV sealed-bid auction described by (q, p)
 - Recall that $q : [0, \infty)^n \rightarrow \Delta(N)$, gives the winning probability in the auction as a function of the vector of bids.
 - $p : N \times [0, \infty)^n \rightarrow \mathbb{R}^n$ gives the payment vector for each player as a function of the identity of the winner and the vector of bids.
- Let $\beta : \mathcal{V} \rightarrow [0, \infty)$ be a monotonically increasing strategy.
- Denote by $m_i(v_i) = m(q, p, \beta; v_i)$ the expected payment that buyer i with private value v_i pays in an auction method (q, p) when all the buyers implement β :

$$m_i(v_i) = \int_{v_{-i} \in \mathcal{V}^{n-1}} \sum_{i_* \in N} q_{i_*}(\beta(v_1), \dots, \beta(v_n)) p_i(i_*, \beta(v_1), \dots, \beta(v_n)) dF_{-i}(v_{-i})$$

Theorem 6.3

Let β be a symmetric and monotonically increasing equilibrium in a sealed-bid auction with independent private values satisfying the following properties:

- 1 the winner of the auction is the buyer with the highest private value.
- 2 the expected payment by a buyer with private value 0 is 0.

Then

$$m_i(v_i) = G_1^{(n-1)}(v_i) \mathbb{E}[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v_i].$$

- Property (1) is called the **efficiency condition**: an efficient auction is one in which, in equilibrium, the object is allocated to the buyer who most highly values it.
- If the seller is willing to sell the object at any nonnegative price, and since the private values of the buyers are nonnegative, an efficient auction **allocates the object with probability 1**.
- What is important about theorem 7.3: the right hand side shows that the expected payment of any bidder **does not depend on the auction method!** (Caveat: assuming the SIPV assumptions and risk-neutrality of the buyers.)

Proof (Thm 6.3) Recall $y_1^{(n+1)} \stackrel{\text{CDF}}{\sim} G_1^{(n+1)}$. Let β be a symmetric & monotonically increasing strategy in auction (q, p) . Let $v_i \in \mathcal{V}$ be the private value of buyer i , $v_i \in (0, \bar{v})$.

If buyer i deviates from the strategy and plays as if her private value is z_i , she wins only if z_i is higher than $y_1^{(n+1)}$, which happens with probability $G_1^{(n+1)}(z_i)$. Her profit is then:

$$U_i(\beta(z_i); \beta_{-i}; v_i) = G_1^{(n+1)}(z_i) v_i - m_i(z_i)$$

Since β is an equilibrium buyer's best reply is $z_i = v_i$.

Next Goal: show that m_i is differentiable and compute its derivative:

Since $U_i(\beta(z_i); \beta_{-i}; v_i)$ attains a maximum at $z_i = v_i$ then we have:

$$v_i G_1^{(n+1)}(v_i) - m_i(v_i) \geq v_i G_1^{(n+1)}(z_i) - m_i(z_i)$$

$$\implies v_i (G_1^{(n+1)}(v_i) - G_1^{(n+1)}(z_i)) \geq m_i(v_i) - m_i(z_i)$$

Exchanging the roles of x_i and z_i we have that:

$$z_i G_i^{(n-1)}(v_i) - m_i(v_i) \leq z_i G_i^{(n-1)}(z_i) - m_i(z_i)$$
$$\Rightarrow z_i (G_i^{(n-1)}(v_i) - G_i^{(n-1)}(z_i)) \leq m_i(v_i) - m_i(z_i)$$

Combining: $z_i (G_i^{(n-1)}(v_i) - G_i^{(n-1)}(z_i)) \leq m_i(v_i) - m_i(z_i) \leq v_i (G_i^{(n-1)}(v_i) - G_i^{(n-1)}(z_i))$

For $z_i \neq v_i$ dividing by $v_i - z_i$ yields:

$$z_i \left[\frac{G_i^{(n-1)}(v_i) - G_i^{(n-1)}(z_i)}{v_i - z_i} \right] \leq \frac{m_i(v_i) - m_i(z_i)}{v_i - z_i} \leq v_i \left[\frac{G_i^{(n-1)}(v_i) - G_i^{(n-1)}(z_i)}{v_i - z_i} \right]$$

Taking limits as $z_i \rightarrow v_i$ yields: $m_i'(v_i) = \lim_{z_i \rightarrow v_i} \frac{m_i(v_i) - m_i(z_i)}{v_i - z_i} = v_i g_i^{(n-1)}(v_i)$
for all $v_i \in (0, \bar{v})$

Note: m_i' is independent of the auction method.

Using the Fundamental Theorem of Calculus and $m_i(0) = 0$ we get

$$m_i(v_i) = m_i(0) + \int_0^{v_i} m_i'(s) ds = \int_0^{v_i} s g_i^{(n+1)}(s) ds$$

$$= G_i^{(n+1)}(v_i) E[Y_i^{(n+1)} | Y_i^{(n+1)} \leq v_i].$$

$$\left\{ \text{Recall: } E[Y_i^{(n+1)} | Y_i^{(n+1)} \leq v_i] = \frac{1}{G_i^{(n+1)}(v_i)} \int_0^{v_i} s g_i^{(n+1)}(s) ds \right\}$$

Q.E.D.

Corollary 6.4 (The Revenue Equivalence Theorem)

In a symmetric-sealed bid auction with independent private values, let β be a symmetric and monotonically increasing equilibrium satisfying the following properties (1) and (2) from theorem 7.3. Then the seller's expected revenue is

$$\pi = n \int_{\mathcal{V}} m_i(v) f(v) dv$$

where

$$m_i(v) = G_1^{(n-1)}(v) \mathbb{E}[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v].$$

Equilibrium Strategy in an All-Pay Auction

Suppose β^{all} is a symmetric, monotonically increasing equilibrium strategy for the All-pay IPV auction. Suppose $\beta^{all}(0) = 0$. We can use the RET to derive the shape of β^{all} .

In an all-pay auction $m_i(v) = \beta^{all}(v)$.

The RET yields $m_i(v) = \int_0^v s g_i^{(n-1)}(s) ds$

$$\Rightarrow \beta^{all}(v) = \int_0^v s g_i^{(n-1)}(s) ds //$$

Verifying it is an equilibrium: Suppose all but i follow β^{all} . If i bids $\beta^{all}(z_i)$ for some value $z_i \in \mathcal{V}$ when her value is v_i , then the expected payoff is: $G_i^{(n-1)}(z_i)v_i - \beta^{all}(z_i) = G_i^{(n-1)}(z_i)v_i - \int_0^{z_i} s g_i^{(n-1)}(s) ds$. Integrating by parts yields:
$$= G_i^{(n-1)}(z_i)(v_i - z_i) + \int_0^{z_i} G_i^{(n-1)}(s) ds \xrightarrow{FOC} g_i^{(n-1)}(z_i)(v_i - z_i) = 0 \quad \therefore v_i = z_i \text{ is optimal.}$$

Equilibrium Strategy in a Third-Price Auction

Suppose that $\beta: \mathcal{V} \rightarrow \mathbb{R}_+$ is a symmetric, monotone increasing eq. bidding function $\beta(0) = 0$. By RET:

$$m_i(v_i) = \int_0^{v_i} s g_1^{(n-1)}(s) ds$$

On the other hand a bidder with value v_i wins the auction when: $\beta(v) > \beta(y_1^{(n-1)}) \Leftrightarrow v_i > y_1^{(n-1)}$ [$v_i > y_1^{(n-1)} > y_2^{(n-1)}$]

Then this buyer pays $\beta(y_2^{(n-1)})$, then the expected payment is:

$$m_i(v_i) = G_1^{(n-1)}(v_i) \cdot \mathbb{E}[\beta(y_2^{(n-1)}) \mid y_1^{(n-1)} \leq v_i]$$

$$g_{1,2}^{(n-1)}(y_1, y_2) = (n-1)(n-2) f(y_1) f(y_2) F(y_2)^{n-3} \quad y_1 \geq y_2$$

$$g_2^{(n-2)}(y_2 \mid y_1^{(n-1)} \leq v_i) = \frac{\Pr(y_1^{(n-1)} \leq v_i, y_2)}{\Pr(y_1^{(n-1)} \leq v_i)} = \frac{\int_{y_2}^{v_i} (n-1)(n-2) f(y_1) f(y_2) F(y_2)^{n-3} dy_1}{G_1^{(n-1)}(v_i)}$$

$$g_2^{(n-2)}(y_2 | Y_1^{(n-1)} \leq v_i) = \frac{\int_{y_2}^{v_i} (n-1)(n-2) f(y_1) f(y_2) F(y_2)^{n-3} dy_1}{G_1^{(n-1)}(v_i)} = \frac{(n-1) g_1^{(n-2)}(y) [F(v_i) - F(y)]}{G_1^{(n-1)}(v_i)}$$

Numerator : $(n-1)(n-2) f(y_2) F(y_2)^{n-3} \int_{y_2}^{v_i} f(y_1) dy_1 = (n-1)(n-2) f(y_2) F(y_2)^{n-3} [F(v_i) - F(y_2)]$
 $\quad \quad \quad = g_1^{(n-2)}$

$$E[\beta(Y_2^{(n-1)}) | Y_1^{(n-1)} \leq v_i] = \int_0^{v_i} \beta(y) \frac{(n-1) g_1^{(n-2)}(y) [F(v_i) - F(y)]}{G_1^{(n-1)}(v_i)} dy$$

$$m_i(v_i) = G_1^{(n-1)}(v_i) \cdot E[\beta(Y_2^{(n-1)}) | Y_1^{(n-1)} \leq v_i] = \int_0^{v_i} \beta(y) (n-1) g_1^{(n-2)}(y) [F(v_i) - F(y)] dy$$

Then from RET :

$$\int_0^{v_i} \beta(y) (n-1) g_1^{(n-2)}(y) [F(v_i) - F(y)] dy = \int_0^{v_i} y g_1^{(n-1)}(y) dy$$

$$\int_0^{v_i} \beta(y) (n-1) g_i^{(n-2)}(y) [F(v_i) - F(y)] dy = \int_0^{v_i} y g_i^{(n-1)}(y) dy$$

Differentiate wrt v_i :

$$\frac{d}{dv_i} \left[F(v_i) \int_0^{v_i} \beta(y) (n-1) g_i^{(n-2)}(y) dy - \int_0^{v_i} \beta(y) (n-1) g_i^{(n-2)}(y) F(y) dy \right] = \frac{d}{dv_i} \left(\int_0^{v_i} y g_i^{(n-1)}(y) dy \right)$$

$$f(v_i) \int_0^{v_i} \beta(y) (n-1) g_i^{(n-2)}(y) dy + \cancel{F(v_i) \cdot \beta(v_i) (n-1) g_i^{(n-2)}(v_i)} - \cancel{\beta(v_i) (n-1) g_i^{(n-2)}(v_i) F(v_i)} = v_i g_i^{(n-1)}(v_i)$$

$$\cancel{f(v_i)} \int_0^{v_i} \beta(y) \cancel{(n-1)} g_i^{(n-2)}(y) dy = v_i g_i^{(n-1)}(v_i) = v_i \cancel{(n-1)} \cancel{f(v_i)} F(v_i)^{n-2}$$

$$\frac{d}{dv_i} \left(\int_0^{v_i} \beta(y) g_i^{(n-2)}(y) dy \right) = \frac{d}{dv_i} \left(v_i F(v_i)^{n-2} \right)$$

$$\begin{aligned} \beta(v_i) g_i^{(n-2)}(v_i) &= F(v_i)^{n-2} + v_i (n-2) F(v_i)^{n-3} f(v_i) \\ &= G_2^{(n-2)}(v_i) + v_i g_i^{(n-2)}(v_i) \end{aligned}$$

$$\begin{aligned}\beta(v_i) g_i^{(n-2)}(v_i) &= F(v_i)^{n-2} + v_i (n-2) F(v_i)^{n-3} f(v_i) \\ &= G_1^{(n-2)}(v_i) + v_i g_i^{(n-2)}(v_i)\end{aligned}$$

$$\beta(v_i) = v_i + \frac{G_1^{(n-2)}(v_i)}{g_i^{(n-2)}(v_i)} = v_i + \frac{F(v_i)^{n-2}}{(n-2) f(v_i) F(v_i)^{n-3}}$$

$$\beta(v_i) = v_i + \frac{F(v_i)}{(n-2) f(v_i)}, \quad \text{for } \beta \text{ to be increasing}$$

a sufficient condition is for $\frac{F(x)}{f(x)}$ to be increasing

$$\Leftrightarrow \ln F \text{ is concave.} \Leftrightarrow F \text{ is log-concave.}$$